

Web-Based Device Controller With Arduino Board



PRIYANKA KUMARI

An Arduino board can also be used to control electrical devices and appliances over the Internet. Just by clicking a few buttons on a webpage, the devices can be operated from a remote location. For this the board needs to be Ethernet-enabled using Ethernet shield. This project illustrates the point by using Wiznet W5100 Ethernet shield to control a servo motor and an LED.

Circuit and working

Fig. 1 shows circuit of the web-based device controller. The circuit is built around Arduino UNO board (board1), Arduino Ethernet shield (shield1) and servo motor (M1).

Servo motor. A servo motor mainly consists of a DC motor, gear system, position sensor (generally a potentiometer) and control electronics as shown in Fig. 2. The DC motor is connected to a gear mechanism, which provides feedback to a position sensor. The potentiometer allows the control circuit to monitor the current angle of the servo motor. If the shaft is at correct angle, the motor shuts off. If the circuit finds that the angle is incorrect, it will keep the motor on until the angle is correct. A normal servo is used to control the angular motion up to 180 degrees.

Angle of rotation. The angle of servo motor rotation is determined by the duration of a pulse that is applied to the control wire. The servo expects a pulse every 20 milliseconds (0.02 seconds). The length of the pulse will determine how far the motor rotates. A 1.5-millisecond pulse, for example, makes the motor turn to the 90-degree position (often called the neutral position). If the pulse is shorter than 1.5ms, the motor turns the shaft lesser than

90 degrees. If the pulse is longer than 1.5ms, the shaft turns closer to 180 degrees as shown in Fig. 3.

Arduino UNO board. Arduino is an open source electronics prototyping platform based on flexible, easy-to-use

hardware and software. It is intended for artists, designers, hobbyists and anyone interested in creating interactive objects or environments.

Arduino UNO is a board based on ATmega328 microcontroller. It consists of 14 digital input/output pins, six analogue inputs, a USB connection for programming the on-board microcontroller, power jack, an ICSP header and a reset button. It is operated with a 16MHz crystal oscillator and contains everything needed to support the microcontroller. It is very easy

to use as the user simply needs to connect it to a computer with a USB cable or power it with an AC-to-DC adaptor or battery to get started. The microcontroller on the board is programmed using Arduino programming language and Arduino development environment.

Arduino Ethernet shield. The Arduino Ethernet shield allows an Arduino board to connect to the Internet. It is based on the Wiznet W5100 Ethernet chip, which provides a network (IP) stack capable of both TCP and UDP. It supports up to four simultaneous socket connections. Use the Ethernet library to write sketches which connect to the Internet using the shield.

The Ethernet shield connects to

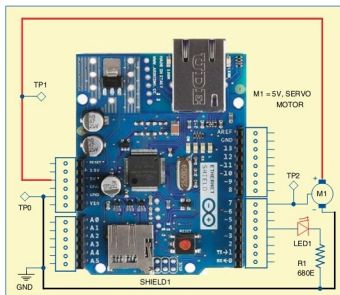


Fig. 1: Circuit diagram of the web-based controller

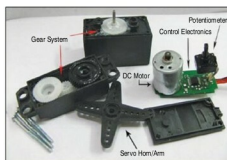


Fig. 2: Servo motor components

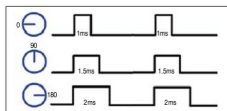


Fig. 3: Servo positions